

MT8002 Elements of Set Theory
7.4 Ordered Sets. Well-Ordered Sets

Andrés Sicard-Ramírez

Universidad EAFIT

Semester 2026-2

Preliminaries

Textbook

Derek Goldrei (1996). Classic Set Theory. A Guided Independent Study.

Convention

The numbers and page numbers assigned to chapters, examples, exercises, figures, quotes, sections and theorems on these slides correspond to the numbers assigned in the textbook.

Outline

Well-Ordered Sets

Some Properties of Well-Ordered Sets

References

Outline

Well-Ordered Sets

Some Properties of Well-Ordered Sets

References

Well-Ordered Sets

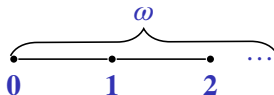
Definition (p. 188)

Let $(X, <)$ be a linearly ordered set. The relation $<$ is a **well-order** (or the set X is **well-ordered** by the relation $<$) if every non-empty subset A of X has a least element, i.e., there is some element $a \in A$ such that $a \leq x$, for all $x \in A$.

Well-Ordered Sets

Example

The canonical example of a well-ordered set is ω , i.e., $(\mathbb{N}, \in)$,



because every non-empty subset of ω has a least element (Theorem 3.9).

Well-Ordered Sets

Example

The closed interval $[0, 1]$ of $\mathbb{R}$ with the usual order is not well-ordered because the interval $(0, 1) \subseteq [0, 1]$ has no least element.

Well-Ordered Sets

Example

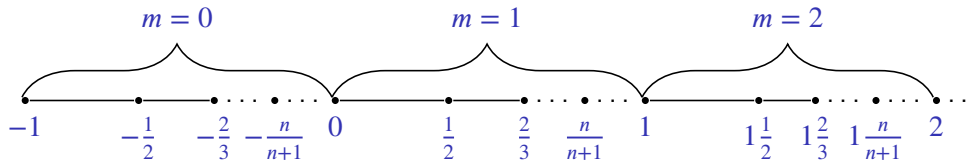
The closed interval $[0, 1]$ of $\mathbb{R}$ with the usual order is not well-ordered because the interval $(0, 1) \subseteq [0, 1]$ has no least element.

Let $a, b \in \mathbb{R}$, with $a < b$. In general, any subset of $\mathbb{R}$ containing an open interval (a, b) is not well-ordered.

Well-Ordered Sets

Example

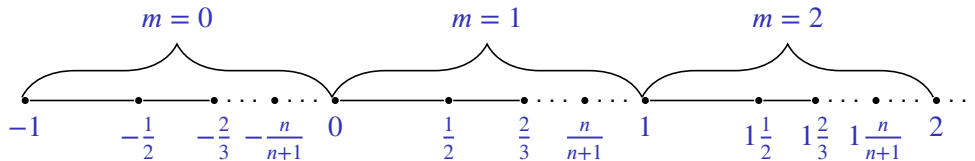
The subset $X = \{m - \frac{1}{n+1} : m, n \in \mathbb{N}\}$ of $\mathbb{Q}$ with the usual order is well-ordered.



Well-Ordered Sets

Example

The subset $X = \{m - \frac{1}{n+1} : m, n \in \mathbb{N}\}$ of $\mathbb{Q}$ with the usual order is well-ordered.



Note that

$$m - \frac{1}{n+1} < m' - \frac{1}{n'+1} \text{ iff } \begin{cases} m < m', \text{ or} \\ m = m' \text{ and } n < n'. \end{cases}$$

(continued on next slide)

Well-Ordered Sets

Example (continuation)

Let A a non-empty subset of X .

“The least element of A has to have the least m available among the elements of A and the smallest n for this m .” (p. 189)

We find the least element of A using the well-order property of natural numbers.

Let $B = \left\{ m \in \mathbb{N} : m - \frac{1}{n+1} \in A, \text{ for some } n \in \mathbb{N} \right\}$ and let m_0 the least element of B .

Let $C = \left\{ n \in \mathbb{N} : m_0 - \frac{1}{n+1} \in A \right\}$ and let n_0 the least element of C .

Then $m_0 - \frac{1}{n_0+1}$ is the least element of A .

Outline

Well-Ordered Sets

Some Properties of Well-Ordered Sets

References

Some Properties of Well-Ordered Sets

Theorem

Let $(X, <)$ be a well-ordered set and let $A \subseteq X$. Then the set A is well-ordered by the restriction of relation $<$ to A , i.e., $(A, <|_{A \times A})$ is a well-ordered set (Exercise 7.44).

Some Properties of Well-Ordered Sets

Example

The set $5 \doteq \{0, 1, 2, 3, 4\}$ is well-ordered by the relation $\in$.

Some Properties of Well-Ordered Sets

Let X be a set. Recall that an infinite descending $\in$ -chain is a sequence $\langle x_n \rangle$ such that $x_{n+1} \in x_n$, for all $n \in \mathbb{N}$, that is,

$$\cdots x_{n+1} \in x_n \in \cdots \in x_2 \in x_1 \in x_0.$$

Some Properties of Well-Ordered Sets

Let X be a set. Recall that an infinite descending $\in$ -chain is a sequence $\langle x_n \rangle$ such that $x_{n+1} \in x_n$, for all $n \in \mathbb{N}$, that is,

$$\cdots x_{n+1} \in x_n \in \cdots \in x_2 \in x_1 \in x_0.$$

Definition (p. 190)

Let $(X, <)$ be an ordered set. An **infinite descending $<$ -chain** is a sequence $\langle x_n \rangle$ such that $x_{n+1} < x_n$, for all $n \in \mathbb{N}$, that is,

$$\cdots x_{n+1} < x_n < \cdots < x_2 < x_1 < x_0.$$

Some Properties of Well-Ordered Sets

Theorem

Let $(X, <)$ be a well-ordered set and suppose that X contains an infinite descending $<$ -chain. Then X is not well-ordered by $<$ (Exercise 7.45).

Some Properties of Well-Ordered Sets

Example

The set $\mathbb{Z}$ with the usual order $<$ is not a well-ordered set because the subset of negative integers forms an infinite descending $<$ -chain.

$$\dots < -3 < -2 < -1$$

Some Properties of Well-Ordered Sets

Theorem

Let X and Y be ordered sets and let $f : X \longrightarrow Y$ be an order-isomorphism. Then if X is well-ordered then so is Y (Exercise 7.46).

Some Properties of Well-Ordered Sets

Example

The set ω^2 is well-ordered because there is an order-isomorphism between it and the subset $\{m - \frac{1}{n+1} : m, n \in \mathbb{N}\}$ of $\mathbb{Q}$ with the usual order.

Some Properties of Well-Ordered Sets

Exercise 7.47

Suppose that X and Y are well-ordered sets. Show that the ordered sets $X + Y$ and $X \times Y$ are both well-ordered.

Outline

Well-Ordered Sets

Some Properties of Well-Ordered Sets

References

References

Derek Goldrei (1996). Classic Set Theory. A Guided Independent Study. Chapman & Hall (cit. on p. 2).